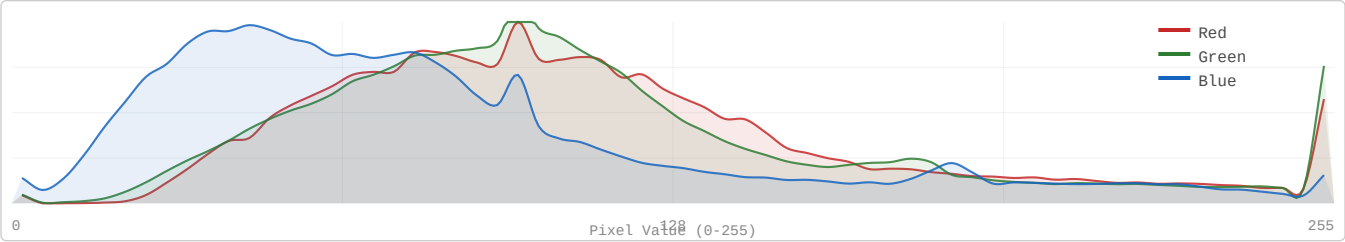
Decorrelation Gap: KODIM13

768x512 | 24-bit RGB | PNG (lossless)

Sunset Sky over Dark Landscape | Near-degenerate (PC1 = 97.6%)

1. RGB Channel Distribution



2. Inter-Channel Correlation: RGB vs YCbCr

RGB (Original)

	R	G	B
R	1.0000	0.9818	0.9469
G	0.9818	1.0000	0.9639
B	0.9469	0.9639	1.0000

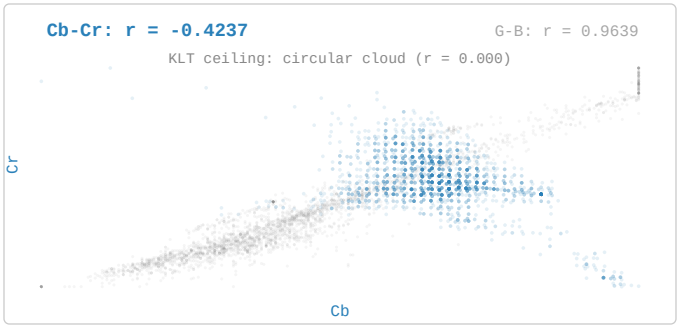
Cb-Cr: -0.424

(KLT ceiling = 0.000)

YCbCr (BT.601)

	Y	Cb	Cr
Y	1.0000	0.0207	-0.1830
Cb	0.0207	1.0000	-0.4237
Cr	-0.1830	-0.4237	1.0000

3. Cb vs Cr Scatter with G-B background overlay



4. Pairwise Decorrelation Gap

Channel Pair	RGB	YCbCr (BT.601)	Residual |r|
R-G → Y-Cb	0.981787	0.020655	0.0207
R-B → Y-Cr	0.946925	-0.182968	0.1830
G-B → Cb-Cr	0.963893	-0.423709	0.4237
Avg |r|	0.964202	0.209110	0.2091

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5. Decorrelation Efficiency & Blue Independence

Decorrelation Efficiency

78.3% High

0% BT.601 leaves all — 100% KLT ceiling

Blue Channel Independence (Baetzel 2026a)

3.4%

0% Fully predictable — 100% Fully independent

6. 4:2:0 Chroma Subsampling — Information Cost

Channel	PSNR (dB)
R	45.81 dB
G	49.11 dB
B	39.57 dB
Overall	43.04 dB

Method: BT.601 RGB→YCbCr | 2x box downsample Cb,Cr | nearest upsample | BT.601 inverse

7. PCA Baseline Reference (Baetzel 2026a)

Eigendecomposition

PC1	97.60%
PC2	1.89%
PC3	0.51%

Eigenvector Loadings

	R	G	B	Eigenvalue
PC1	-0.5614	-0.5821	-0.5882	8367.17
PC2	-0.5688	-0.2449	0.7852	161.63
PC3	-0.6011	0.7754	-0.1936	44.04

8. PCA Derived Metrics

Condition Number

189.99

PC1/PC2: 51.77 | PC2/PC3: 3.67

Blue Channel Independence

3.4%

Variance not captured by PC1

PC1 Dominant Channel

B

Loading: B (0.59)

9. Redundancy Profile

Classification: Near-degenerate | CN: 189.99 | PC1: 97.6%

RGB Correlation: Avg |r| = 0.964202 | R-G: 0.9818 | R-B: 0.9469 | G-B: 0.9639

YCbCr Residual: Avg |r| = 0.209110 | Y-Cb: 0.0207 | Y-Cr: -0.1830 | Cb-Cr: -0.4237

Efficiency: 78.3% of KLT ceiling | Gap: 0.2091 avg |r| remaining

Blue Channel: 3.4% independent | 4:2:0 PSNR: 43.04 dB

References

[1] Baetzel, J. (2026a). "Statistical Characterization of Inter-Channel Redundancy Structure"
[2] Wallace, G.K. "The JPEG still picture compression standard." IEEE Trans. CE 38(1), 1992.
[3] ITU-R Recommendation BT.601: Studio Encoding Parameters, 1982.
[4] Watanabe, S. "Karhunen-Loeve Expansion and Factor Analysis," pp. 635-660, 1965.